

Reviewer Pack — Minimal Tensor Rank of Entanglement Measures for DISJOINTNES...

Entry b453ebc39686 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Tensor Rank of Entanglement Measures for DISJOINTNESS

Entry ID: b453ebc39686 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-21 18:31:03 UTC

1. Conjecture

Field A (mathematical branch): Free Probability Theory **Field B** (complexity object): Communication Complexity of DISJOINTNESS

Statement:

[‘For any entangled state $|\psi\rangle$ in a Hilbert space of dimension n , the minimal tensor rank $\tau(|\psi\rangle)$ of any representation of $|\psi\rangle$ is lower bounded by the randomized communication complexity $CC_DISJ(n)$, i.e., $\tau(|\psi\rangle) = \Omega(CC_DISJ(n))$.’, ‘This bound holds for all possible representations of the entangled state.’, ‘The statement holds if and only if there exists a polynomial-time algorithm that can determine the minimal tensor rank of any given entangled state.’]

Rationale (proposer’s reasoning):

[‘Free probability theory provides tools to analyze quantum systems with entanglement. The concept of minimal tensor rank has been used in entanglement studies, but its connection to communication complexity is unexplored.’, ‘This conjecture bridges free probability theory and communication complexity by suggesting a lower bound on the randomized communication complexity of disjointness using entanglement measures.’, ‘If proven true, this would provide a new approach to understanding the complexity of disjointness problems through quantum entanglement.’]

Taxonomy category: LIFTING (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): bca457d49425af36

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all 30 random seeds, the computed minimal tensor rank $\tau(|\psi\rangle)$ correlates positively with $CC_DISJ(n)$ at a p-value ≤ 0.05 , and the mean correlation coefficient across all seeds exceeds 0.5.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "free probability theory" AND "communication complexity of disjointness" AND "minimal tensor rank" - "entanglement measures" AND "randomized communication complexity" AND "tensor rank" - "algorithm for determining minimal tensor rank" AND "free probability theory" AND "disjointness communication complexity"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_string(length):
        return ''.join(random.choice('01') for _ in range(length))

    def create_entangled_state(X, Y):
        n = len(X)
        entangled_state = {}
        for i in range(n):
            if X[i] == '1' and Y[i] == '1':
                entangled_state[(i, i)] = 1
        return entangled_state

    def compute_tensor_rank(entangled_state):
        # Simplified version of tensor rank computation
        rank = len(entangled_state)
        return rank

    def compute_randomized_communication_complexity(n):
        # Simplified version of CC_DISJ(n)
        return n * (n - 1) // 2

    n_values = [5, 10, 15, 20, 30, 40]
    correlation_coefficients = []
```

```

for n in n_values:
    X = generate_random_string(n)
    Y = generate_random_string(n)
    entangled_state = create_entangled_state(X, Y)
    tensor_rank = compute_tensor_rank(entangled_state)
    cc_disj = compute_randomized_communication_complexity(n)

    if tensor_rank == 0 or cc_disj == 0:
        continue

    correlation_coefficients.append(tensor_rank / cc_disj)

if not correlation_coefficients:
    return {
        "metric_name": "correlation_coefficient",
        "metric_value": None,
        "instances_tested": len(correlation_coefficients),
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

mean_correlation = sum(correlation_coefficients) / len(correlation_coefficients)
return {
    "metric_name": "correlation_coefficient",
    "metric_value": mean_correlation,
    "instances_tested": len(correlation_coefficients),
    "conjecture_holds": mean_correlation > 0.5,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) /
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    else:
        for r in results:
            if not r["conjecture_holds"]:
                counterexample = f"n={r['instances_tested']}, tensor_rank={r['metric_value']}"
                print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={seed}")
                break

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
{'metric_name': 'correlation_coefficient', 'metric_value': 0.07023080099305144, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.04563970344369619, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.03872865556168641, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.06346393909914781, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.05023736766023518, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.046796256778107954, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.05033287610420097, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.01786725877469979, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.04004660560921904, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.058445516975462526, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.05826509982589837, 'instances_tested': 15}
TRIAL: {'metric_name': 'correlation_coefficient', 'metric_value': 0.05542568106088977, 'instances_tested': 15}
RESULT: FALSIFIED counterexample="n=6, tensor_rank=0.033716659760216934" first_failing_seed=929
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The conjecture has been tested on only a small number of instances ($n \leq 15$). This is insufficient to establish the validity of the conjecture, as it may not scale with n and could be an artifact of the limited sample size.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results show that for at least one instance ($n=6$), the computed minimal tensor rank $\tau(|\psi\rangle)$ does not correlate positively with $CC_DISJ(n)$, as | next: Further investigation is needed to determine if the conjecture holds for larger values of n . Additional tests with a wider range of entangled states and dimensions should be conducted to validate or falsify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 12

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14849
2	propose	ollama_remote	glm4:latest	0	0	13158
3	propose	ollama_remote	glm4:latest	0	0	13833
4	preregistration	ollama_remote	glm4:latest	0	0	9201
5	novelty	ollama_remote	glm4:latest	0	0	8714
6	novelty	ollama_remote	glm4:latest	0	0	9038
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17462
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13988
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8685
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8946

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
11	critic	ollama_remote	glm4:latest	0	0	12756
12	judge	ollama_remote	glm4:latest	0	0	9981

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 140612 ms total latency. Provider mix: {'ollama_remote': 12}

(full prompt+response transcripts available in research/audit/b453ebc39686.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/b453ebc39686.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/b453ebc39686.tar.gz (if generated)